National Chung Hsing University

Graduate Institute of Data Science and Information Computing

Master Thesis

Analyzing Improvement in Prediction of Residential Short-Term Rental Prices in Dallas, Texas Based on Localized Violent Crime Using Gradient Boosting Models

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Abstract

This thesis examines whether localized violent-crime exposure improves the prediction of residential short-term rental prices in Dallas, Texas, and whether improvement is robust to spatial evaluation design. Using listing data from 5,290 Airbnb properties and 28,136 violent crime incidents recorded by the Dallas Police Department between 2023 and 2025, the study constructs listing-centered crime exposure variables at five buffer distances: 100, 200, 300, 500, and 1,000 meters. The analytical framework combines hedonic regression benchmarks with gradient boosting machine (GBM) models principal component analysis, and random forest models, evaluated under a conventional random train-test split and spatial block cross-validation.

A grid search across 48 hyperparameter combinations using 5-fold cross-validation identifies the optimal GBM configuration. Under random splitting, the crime-enhanced GBM achieves a test R^2 of 0.701, a modest improvement over the baseline model's 0.689. SHapley Additive exPlanations (SHAP) values confirm that structural features dominate the pricing signal, with crime variables ranking 8th out of 15 in feature importance. However, under spatial block cross-validation, the apparent improvement disappears. Both models achieve a mean spatial CV R^2 of 0.574. This indicates that the crime contribution observed under random splitting is largely an artifact of spatial autocorrelation between training and test observations.

Hedonic regression benchmarks separately confirm that violent crime is negatively and significantly associated with listing price across all buffer specifications, with the strongest per-incident effect at the 100-meter scale. SHAP dependence analysis reveals a scale-dependent pattern in which only the 100-meter buffer contributes negatively to predicted price, while broader buffers (200m, 300m, 500m, 1,000m) contribute positively on average for entire-home listings.

These results carry two implications. Local violent crime is a meaningful neighborhood disamenity in Dallas's short-term rental market but its predictive influence is concentrated at the most immediate spatial scale. Methodologically, the predictive contribution of spatially correlated features can be substantially overstated by random train-test splits, and spatial block cross-validation should be treated as a standard robustness check rather than an optional diagnostic in urban prediction studies.

Keywords: short-term rental pricing; violent crime; gradient boosting; SHAP values; spatial cross-validation; hedonic pricing; Dallas, Texas; random forest

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Chapter 1:

Introduction

1.1 Overview and Motivation

Urban neighborhoods are shaped by many factors, including physical infrastructure, accessibility, housing characteristics, development intensity, and public safety. Among these, crime remains one of the most visible indicators of neighborhood quality. Crime influences how residents, visitors, investors, and city planners perceive urban space (Taylor, 1995; Tita, Petras, & Greenbaum, 2006). A neighborhood that is known for robbery, assault, or homicide may be viewed as less desirable and riskier even when the housing stock is otherwise attractive and well located. For this reason, urban research has treated crime as a negative neighborhood externality that affects economic value. The concern is that crime changes what a location is worth to everyone else who might consider living, visiting, or investing there.

This thesis examines whether localized violent-crime exposure improves the prediction of residential short-term rental prices in Dallas, Texas. Instead of focusing on traditional home sale values, the study uses residential short-term rental listing prices as the market outcome of interest. Digital accommodation platforms such as Airbnb have transformed residential units into short-term market assets that can be priced dynamically and observed at a spatial scale (Guttentag, 2015; Barron, Kung, & Proserpio, 2021). Unlike owner-occupied housing, where prices are only observed at the moment of a transaction, residential short-term rental prices can adjust frequently in response to neighborhood conditions, competition, and demand. As a result, short-term rental prices are a useful proxy for examining whether local violent crime contributes meaningful information about urban price variation. The question being asked is whether local violent crime contains useful information for forecasting prices that standard listing features do not already capture. Asking this question in predictive rather than purely explanatory terms is grounded in what a working model needs to do (Shmueli, 2010; James, Witten, Hastie, & Tibshirani, 2021).

There is also a methodological motivation for this study that goes beyond the Dallas case. Machine learning models for urban property pricing rely on training-test splits to evaluate performance, but they do not always account for the fact that listings located close together in space tend to share similar neighborhood conditions. This spatial autocorrelation creates a form of information leakage when random splits are used. A training observation placed next to a test observation can share very similar crime counts, which allows the model to capture geographic patterns rather than learning a crime-price relationship that generalizes across the city. This thesis addresses that concern directly by implementing spatial block cross-validation alongside conventional evaluation. Both results are reported and compared, which allows the study to separate genuine predictive improvement from an artifact of spatial proximity in the data.

1.2 Study Area

Dallas, Texas is an appropriate case study for this research for several reasons. As one of the largest metropolitan centers in the United States with a population of more than 8 million, Dallas contains substantial variation in neighborhood form, socioeconomic conditions, development intensity, and crime exposure. It also supports an active residential short-term rental market that is distributed across multiple parts of the city instead of a single tourist district. The prediction model needs meaningful differences across listings and neighborhoods to learn a relationship between crime and price. If all listings faced similar safety conditions, there would be no variation in crime exposure for the model to detect.

Publicly available police incident data and geocoded short-term rental listing data make it possible to measure neighborhood safety at a localized scale. The economic effects of neighborhood safety are often felt at the level of blocks and immediate surroundings rather than in entire administrative units (Kim, Phipps, & Anselin, 2003; Ratcliffe, 2010). Dallas is also not dominated by a single type of housing or a single tourism economy. Listings are found in dense urban cores, transitional neighborhoods, and more suburban residential areas, each with different safety profiles and pricing structures. This range of neighborhood types makes it possible to examine how crime and price interact across a variety of urban settings within one city.

The 2023 to 2025 observation window was deliberately chosen. The thesis uses only incidents from this recent period to reflect current neighborhood conditions rather than historical patterns that may no longer apply. The beginning of the crime window in 2023 also avoids the residual effects of reduced public activity during the COVID-19 pandemic period, when the reported levels of crime in many cities were lower relative to normal urban conditions. Using this window ensures that the safety features in the model reflect a normal post-pandemic urban environment in Dallas. It is also worth noting that Dallas is not a resort city or a convention-dominated market where the majority of short-term rental demand comes from tourism. This means that price variation across listings is more likely to reflect genuine neighborhood-level differences in desirability, safety, and local character. In a resort city, much of the pricing signal might be absorbed by proximity to the resort itself, leaving little room for neighborhood safety to matter. In Dallas, the residential short-term rental market is more distributed and more sensitive to the heterogeneity of neighborhood conditions across the city, which is where crime exposure is most likely to have predictive value.

1.3 Research Questions

The central research question is:

> Does localized violent-crime exposure improve the prediction of residential short-term rental prices in Dallas, Texas?

This question is supported by five secondary questions:

1. Does the inclusion of violent-crime buffer features improve predictive performance relative to a baseline model built only on listing and neighborhood characteristics?
2. Which spatial scale of violent-crime exposure—100 meters, 200 meters, 300 meters, 500 meters, or 1,000 meters—is most useful for prediction?
3. How does a crime-enhanced gradient boosting model perform relative to benchmark regression models and a PCA-based alternative?
4. Which variables contribute most strongly to price prediction, and where do crime-related variables rank relative to structural and neighborhood predictors as measured by SHAP values?
5. Does the crime contribution survive spatial block cross-validation, which prevents spatial leakage between training and test observations?

These questions make the study more specific than a broad claim that crime lowers prices. The thesis evaluates whether violent crime improves price prediction in a Dallas-specific residential short-term rental market, and it tests whether any improvement holds up under a more demanding approach that accounts for spatial structure.

1.4 Contribution

This thesis makes several contributions to existing research. By treating violent crime as a spatial predictive feature, it extends the urban externalities literature into a machine learning prediction context. This is different from prior hedonic studies that focus on establishing significance of a crime coefficient. The discovery here is whether crime improves out-of-sample prediction, which is a harder and more practical test. A feature that is statistically significant in regression is not necessarily one that improves prediction on new data, and the distinction matters for anyone who wants to build a pricing tool rather than estimate a structural model.

By using residential short-term rental listing prices as the market outcome, the study adds to a literature that has focused primarily on tourism, structural listing attributes, and platform dynamics. The thesis combines police incident data, geographic projection, buffer-based spatial joins, gradient boosting, SHAP decomposition, and spatial block cross-validation in a configuration that has not previously been applied to short-term rental pricing. Each component of this pipeline is individually straightforward, but the combination has not been applied in this way to the residential short-term rental pricing context before.

The spatial cross-validation finding is the most important methodological contribution of the thesis. Under conventional random splitting, the crime-enhanced model improves R^2 from 0.689 to 0.701. Under geographic holdout, that improvement disappears entirely, with both models reaching a mean spatial cross-validation R^2 of 0.574. This result suggests that the predictive contribution of spatially correlated features may be substantially overstated when evaluated using only random splits, and warrants verification in other settings. Based on this finding, the thesis recommends that spatial block cross-validation be incorporated as a standard robustness check in urban prediction studies rather than treated as an optional diagnostic.

The remainder of this thesis is organized as follows. Chapter 2 reviews the relevant literature on crime as a neighborhood externality, hedonic pricing theory, short-term rental market research, spatial exposure measurement, and predictive modeling. Chapter 3 explains the research design, data sources, spatial processing steps, feature engineering approach, and modeling strategy. Chapter 4 presents the descriptive overview, regression benchmark results, machine learning model comparisons, SHAP feature importance, and spatial cross-validation findings. Chapter 5 discusses the meaning of the results, the limitations of the approach, directions for future research, and the broader implications for how urban price prediction studies should be evaluated.

Chapter 2:

Literature Review

2.1 Overview and Organization

Urban neighborhood research has long examined how local externalities influence market value, residential behavior, and perceptions of place. This thesis builds on that tradition and applies it to the residential short-term rental market in Dallas, Texas. Specifically, the study asks whether localized violent-crime exposure improves the prediction of residential short-term rental prices when added to standard listing and neighborhood features. To justify this question, the literature review brings together several related bodies of research: crime as a neighborhood disamenity, hedonic pricing theory, short-term rental price formation, spatial approaches to measuring local crime exposure, and predictive modeling.

This chapter is organized into five sections. Section 2.2 reviews the literature on crime as a negative neighborhood externality and explains why violent crime is a particularly useful measure of perceived safety risk. Section 2.3 examines hedonic pricing theory and short-term rental market research. Section 2.4 reviews spatial approaches to measuring local crime exposure and discusses predictive modeling as a methodological extension of traditional urban price analysis. Section 2.5 identifies the specific gap this thesis addresses and explains how the Dallas study fills it.

2.2 Crime, Neighborhood Safety, and Urban Value

Crime has long been treated as a negative externality in urban research because its effects extend beyond the immediate victim of any single incident. Crime changes how neighborhoods are perceived, how residents and visitors behave, and how market participants assign value to place. Taylor (1995) argues that crime affects communities not only through direct victimization but also through fear, avoidance behavior, and reduced confidence in neighborhood quality. These broader effects suggest that crime should not only be understood as a criminal justice issue. It is also a place-based condition with economic implications. When a neighborhood becomes associated with violence or public disorder, the willingness of people to live, visit, or invest there changes, and that change is reflected in market behavior.

This relationship between crime and urban value is especially important in housing and land market research. From a hedonic perspective, crime enters the price function as a negative attribute, reducing the marginal willingness to pay for a given location. Safety is a neighborhood characteristic, and when that characteristic worsens, willingness to pay may fall. This is consistent with the broader urban economics tradition, where prices reflect both internal dwelling characteristics and external neighborhood conditions (Rosen, 1974; Gibbons, 2004).

Tita, Petras, and Greenbaum (2006) provide relevant evidence by examining how crime affects housing prices at the neighborhood level. Their work shows that crime is economically meaningful and spatially uneven, and that it should be measured in ways that reflect local urban geography. A listing in one part of a district may face a very different safety environment from a listing elsewhere in the same district. Building on this, the present study measures crime at an even finer scale than neighborhood-level work, using listing-centered buffers. Using a single city or district average would conceal the local variation that makes crime a potentially meaningful predictive feature at all.

The distinction between violent and nonviolent crime is also central to this thesis. Not all crime types carry the same meaning for neighborhood perception. Violent offenses such as robbery, assault, and homicide are more directly linked to perceived personal danger than many lower-level property or administrative offenses. Prior hedonic work supports a stronger link between violent crime specifically and housing prices compared to property crime (Gibbons, 2004; Linden & Rockoff, 2008). If the mechanism being studied is local safety and willingness to pay, then violent crime is a more appropriate signal than total reported crime. In urban settings, violent crime is also more visible in public discourse and more strongly tied to the reputational meaning of a place. This makes it a better candidate for a neighborhood-risk feature in a predictive model compared to using all crime types indiscriminately (Taylor, 1995).

This issue is particularly relevant for residential short-term rentals. Permanent residents may accumulate detailed local knowledge about their neighborhood over time, but short-term guests usually do not have this knowledge. They often rely on broad impressions, platform reviews, maps, and general neighborhood reputation when choosing where to stay. For this reason, violent crime is likely to matter both directly and indirectly. Even if a guest does not know the exact incident history of a block, areas associated with violent crime may still be perceived as less attractive or less secure. This perception can affect the prices that hosts are able to charge and the demand that certain listings are able to generate, which is precisely why violent crime exposure belongs in a model of residential short-term rental prices (Lepp Gibson, 2008).

The literature does not support a simplistic view in which crime alone determines price. Urban value is always shaped by tradeoffs, so a neighborhood with elevated violent crime may still maintain relatively high prices if it also offers exceptional centrality, access to nightlife, or proximity to amenities. This is why the present thesis does not treat crime as the sole or dominant explanatory force in the model. Violent crime is treated as one variable within a broader predictive system that includes structural, reputational, and spatial features. The results are expected to show that crime matters alongside these more prominent features rather than as a replacement for them. This framing is consistent with how the existing literature treats crime as a meaningful component of neighborhood quality rather than as a single overriding determinant of market value (Gibbons, 2004; Tita, Petras, & Greenbaum, 2006).

Three key ideas from this body of literature are central to the thesis design. First, crime is a negative neighborhood externality with demonstrable economic implications in urban markets. Second, violent crime is especially relevant when the focus is on local safety and perceived personal risk, which is a more appropriate lens than total crime for a study of guest accommodation choices. Third, local spatial variation matters, which means crime exposure should be measured at a localized scale rather than through broad citywide or district-wide rates that conceal the within-neighborhood differences that drive individual listing pricing.

2.3 Hedonic Pricing and Residential Short-Term Rental Markets

The main theoretical foundation for price studies in urban economics is hedonic pricing theory. Rosen (1974) argued that the price of a differentiated good reflects the combined value of its characteristics. In housing markets, those characteristics include structural traits such as size, number of rooms, and build quality, as well as neighborhood and environmental features such as accessibility, school quality, amenities, and safety (Sirmans, Macpherson, & Zietz, 2005). Hedonic theory is directly relevant to this thesis because it explains why residential short-term rental prices should be understood as bundle outcomes rather than isolated listing decisions. A listing is priced as a combination of what it offers physically and where it is located, including what that location means in terms of safety and neighborhood quality.

Hedonic pricing theory was developed in the context of conventional housing markets but it still applies to short-term rental markets. Short-term rental prices are more dynamic than traditional home sale prices since houses are sold infrequently, while listings can be repriced frequently in response to demand, competition, and local conditions. While the present study uses a cross-sectional snapshot of listing prices rather than panel data, the dense spatial coverage of short-term rental listings still makes them a useful setting for studying how local conditions vary across an urban market (Guttentag, 2015; Barron, Kung, & Proserpio, 2021).

The short-term rental literature supports the use of hedonic logic in platform markets. Chen and Xie (2017) show that Airbnb listing prices respond to listing characteristics, host attributes, and locational factors, which is the kind of multi-dimensional pricing structure that hedonic theory predicts. Teubner, Hawlitschek, and Dann (2017) also show that platform pricing depends on both physical traits and reputation-related signals. Together, these studies establish that residential short-term rental prices are structured outcomes shaped by multiple features, not just posted amounts. This justifies including a range of listing and neighborhood variables in the baseline model before testing whether crime adds further predictive value.

The distinction between residential short-term rentals and hotels is also worth noting here. Hotels are often clustered in commercial or convention-oriented locations that are somewhat insulated from local residential neighborhood conditions. Residential short-term rentals are embedded in neighborhood space (Guttentag, 2015). A guest renting an apartment, house, or private room is temporarily purchasing access to a neighborhood. This means that neighborhood-level variables, including local safety, are especially relevant in residential short-term rental pricing. The surrounding conditions matter more to the guest experience and therefore to the price a host can charge compared to a hotel guest who is more isolated from local neighborhood dynamics.

2.4 Spatial Analysis, Safety, and Predictive Modeling

A major challenge in this thesis is deciding how to measure neighborhood safety. Broad neighborhood or district averages are convenient and easy to compute, but they do not necessarily reflect the safety of a local urban space at the level it is actually experienced by residents and guests. Spatial research shows that urban externalities are unevenly distributed and can be sensitive to the scale at which they are measured, a phenomenon described by the modifiable areal unit problem (MAUP) (Openshaw, 1984; Fotheringham & Wong, 1991). Ratcliffe (2010) emphasizes that crime is fundamentally spatial and that different spatial units can produce very different interpretations of the same underlying data. A listing-level price model needs a localized measure of crime exposure rather than citywide or district-level averages if the goal is to capture the variation that matters for individual listings.

This is where buffer-based measurements become important. The thesis constructs violent-crime exposure variables using listing-centered buffers at multiple distances. The buffer approach has a substantial methodological precedent in spatial criminology and the hedonic crime literature (Linden & Rockoff, 2008; Chica-Olmo, Cano-Guervos, & Tamaris-Turizo, 2019). A 100-meter buffer captures the immediate surroundings of a listing. Larger buffers at 200, 300, 500, and 1,000 meters capture progressively broader local environments. This approach is consistent with spatial externalities literature because it allows exposure to vary at the level of the individual listing rather than treating all listings in a district as facing identical conditions (Kim, Phipps, & Anselin, 2003). Two listings in the same administrative neighborhood can receive very different crime counts if one sits near a high-incident corner while the other is on a quieter residential block.

The predictive framing of the thesis builds on a distinction made by Shmueli (2010) between explanation and prediction. This thesis uses regression benchmarks to establish the explanatory significance and direction of the crime-price relationship, but the main analytical framework is gradient boosting evaluated on held-out test data. For the predictive question this thesis poses, model performance on unseen listings matters more than coefficient significance in a regression table.

Gradient boosting is the appropriate approach here because residential short-term rental pricing is unlikely to be fully linear. The value of a listing may depend on nonlinear combinations of size, property type, neighborhood, reputation, and local crime exposure in ways that standard regression cannot accommodate without manual specification of interaction terms (Friedman, 2001; Hastie, Tibshirani, & Friedman, 2009). Gradient boosting learns these patterns automatically from the data. The specific implementation used is XGBoost (Chen & Guestrin, 2016), which has been shown to perform well on tabular data with mixed feature types.

There is also a methodological concern specific to spatial prediction that makes spatial cross-validation a necessary part of the evaluation framework. When listings are distributed across space and crime features are constructed from nearby incidents, observations that are close together in space will naturally share similar crime counts. This spatial autocorrelation means that a randomly split training and test set will place training observations spatially adjacent to many test observations, allowing the model to effectively use geographic proximity as a proxy for crime exposure during evaluation. The result is an overestimate of how well crime features generalize to genuinely new geographic areas. Spatial block cross-validation addresses this by constructing folds that are geographically separated, forcing the model to predict across space rather than within it (Roberts et al., 2017; Valavi, Elith, Lahoz-Monfort, & Guillera-Arroita, 2019). Including this evaluation approach is not standard practice in the short-term rental pricing literature, and demonstrating its importance is one of the methodological contributions of this thesis.

2.5 Research Gap

The existing research strongly supports the framework of this thesis, but it also highlights a clear gap. Urban studies have shown that crime is a meaningful neighborhood externality with economic consequences. Hedonic studies have shown that price reflects both structural and environmental characteristics. Short-term rental research has shown that listing prices are shaped by structure, reputation, and location. Spatial methods have shown that local externalities should be measured geographically rather than through abstract averages. Predictive modeling literature has shown that test-sample accuracy can be used to evaluate whether the addition of a feature was useful.

What remains is a methodological gap that the combination of these streams reveals. Urban prediction studies routinely use random train-test splits to evaluate spatially correlated features, but the spatial autocorrelation literature suggests this approach may overstate predictive contributions (Roberts et al., 2017; Valavi et al., 2019). No study has tested this concern in a short-term rental pricing context using localized crime exposure. Much of the crime-and-value literature focuses on conventional housing markets rather than platform-based short-term rental markets (Gibbons, 2004; Tita, Petras, & Greenbaum, 2006). Much of the short-term rental literature examines structural features and host characteristics without incorporating fine-grained violent-crime exposure (Chen & Xie, 2017; Teubner, Hawlitschek, & Dann, 2017). This thesis fills that gap by combining listing-level residential short-term rental data, localized violent-crime buffers at multiple spatial scales, regression benchmarks, and machine learning in one applied urban case, with explicit attention to spatial evaluation design.

The next chapter builds on this literature by explaining the data sources, spatial processing steps, feature engineering approach, model structure, and evaluation framework used in the Dallas analysis.

Chapter 3:

Methodology

This chapter explains the full methodological design used to predict residential short-term rental prices in Dallas, Texas, with a specific focus on the contribution of localized violent-crime exposure. The central aim is to determine whether adding violent-crime features to a standard listing-level prediction model improves predictive accuracy, and whether any improvement holds up under spatial cross-validation. Section 3.1 explains the research design and justifies the predictive framing. Section 3.2 describes the two primary data sources and the preparation steps applied to each. Section 3.3 explains the spatial processing steps and feature engineering approach used to construct the crime exposure variables. Section 3.4 describes the full modeling strategy, including the regression benchmarks, the gradient boosting models, the hyperparameter tuning procedure, the spatial block cross-validation design, SHAP feature importance, and the evaluation metrics.

3.1 Research Design

The study uses a cross-sectional spatial predictive design. The unit of analysis is the individual residential short-term rental listing. Each listing is associated with structural characteristics (bedrooms, bathrooms, beds, accommodates, room type) along with platform-derived variables (review score, number of reviews, availability across 365 days). Each listing is also assigned localized violent-crime exposure measures derived from police incident data. These variables together form the feature matrix used to predict nightly listing price.

The design is predictive rather than purely explanatory. A standard explanatory study would ask whether crime has a statistically significant negative coefficient in a regression model and what the direction and magnitude of that effect implies. This thesis includes regression benchmarks that serve this purpose, but the main aim is to evaluate whether crime improves out-of-sample prediction. Following Shmueli (2010), the value of a variable in a predictive study is judged by whether it improves model performance on unseen data rather than by whether it is statistically significant on the training sample.

The design is also spatial. This spatial dimension is central to the thesis because it reflects the idea that neighborhood safety is experienced locally rather than citywide. A listing's pricing environment is shaped by the immediate block, nearby streets, and surrounding activity more than by a broad administrative unit. Constructing listing-specific crime measures rather than relying on district averages preserves this local variation and makes the analysis more sensitive to the real spatial structure of crime and price across Dallas.

Three hypotheses motivate the analysis. First, standard listing characteristics such as bedrooms, bathrooms, room type, and neighborhood will explain a substantial share of price variation, because prior hedonic research consistently shows that structural features dominate listing prices. Second, localized violent-crime features will improve prediction beyond those baseline attributes by a modest amount, because crime is expected to be a secondary contributor to price rather than a primary one. Third, the spatial block cross-validation results may not replicate the improvements observed under random splitting, because spatial autocorrelation in the data could inflate the apparent value of crime features when random splits are used. Testing all three of these hypotheses is what gives the thesis its structure and its contribution.

3.2 Data Sources and Preparation

3.2.1 Residential Short-Term Rental Data

The short-term rental dataset provides the dependent variable and the baseline predictors. Using data from Inside Airbnb (2026), the analytical sample contains 5,290 listings used in the regression benchmarks. A smaller subsample of 4,604 listings is used in the gradient boosting models because of listwise deletion on review-related variables. The dependent variable is nightly listing price, which is parsed from string format and converted to a continuous numeric variable. Observations with missing or zero prices are excluded. The price distribution is right-skewed, which is typical of short-term rental markets. The study winsorizes prices at the 99th percentile, corresponding to $1,581, before applying a log transformation. Log price is used as the dependent variable in all models because it reduces the influence of extreme values and improves model fit. Structural variables including bedrooms, bathrooms, beds, and accommodates are converted from character or factor format into continuous numeric form. Review scores and number of reviews are retained as platform reputation signals. Availability across 365 days is included as a host strategy indicator.

3.2.2 Police Incident Data

The Dallas Police Department incident dataset covers 2023 to 2025 and contains 408,978 total records obtained from the Dallas Open Data Portal (Dallas Police Department, 2026).

Violent crime incidents are filtered using offense descriptions matching the patterns ASSAULT, ROBBERY, MURDER, HOMICIDE, KIDNAP, RAPE, and AGGRAVATED. After applying the offense filter and removing incidents with missing or zero coordinates, 28,136 records remain. The 2023 start date was chosen specifically to avoid the residual effects of reduced public activity during the COVID-19 pandemic period, ensuring that the crime data reflect a post-pandemic urban environment where patterns had returned to typical levels.

This filtering choice is theoretically motivated. The purpose of the crime variable is to capture local safety risk in a way that is meaningful for residential short-term rental pricing. Lower-level property offenses, administrative violations, and minor incidents are less likely to influence how a prospective guest perceives the safety of a neighborhood than violent crimes that carry a direct risk of personal harm. Including all incident types without distinction would introduce noise into the model and potentially dilute the signal from the offenses most directly related to perceived personal safety. This filter is broader than the strict UCR definition of violent crime, since it includes kidnapping and simple assault in addition to murder, rape, robbery, and aggravated assault, to capture the wider range of incidents likely to affect perceived neighborhood safety. The remaining records are converted into point geometries and projected into the same coordinate system as the listing data before any spatial joins are performed.

3.3 Spatial Processing and Feature Engineering

Before any distance-based operations can be performed, both the listing data and the violent-crime incident data must be projected into a common local coordinate system. Raw geographic coordinates expressed in latitude and longitude are angular measurements rather than planar distances. Computing distances directly from latitude and longitude produces distorted results because the relationship between degrees and meters varies with latitude. To address this, both layers are projected into the Texas North Central coordinate system (EPSG:2276), which uses US survey feet as its linear unit. Buffer distances specified in meters are converted into feet using the units package in R before each spatial join. This step is methodologically important because spatial research that performs distance-based analysis in unprojected coordinates introduces systematic measurement error that affects both buffer assignments and spatial joins.

Once both layers are projected into the same coordinate system, circular buffers are drawn around each listing. These buffers define the spatial neighborhoods within which violent crime incidents are counted and assigned to the listing. The thesis constructs five buffer zones at distances of 100, 200, 300, 500, and 1,000 meters. Rather than assuming that one spatial scale is correct, the study compares all five distances in the regression benchmarks and includes all five as features in the crime-enhanced gradient boosting model. For each buffer, the number of violent crime incidents whose coordinates fall within the radius is counted, and this count becomes the crime exposure variable for that listing at that spatial scale. The variables are labeled crime_density throughout for consistency with the analysis code, though strictly speaking they are incident counts rather than densities per unit area. Because the buffers are nested (the 1,000m buffer contains the 500m buffer, and so on), the resulting variables are correlated by construction, which is relevant when interpreting their individual contributions.

The baseline feature set includes continuous structural variables (bedrooms, bathrooms, beds, accommodates, review score, number of reviews, availability) and categorical variables (room type, neighborhood). All factor levels are encoded across the full dataset before the train-test split to prevent factor-level mismatches between training and holdout subsets. This is a structural encoding step only so no statistical parameters are learned from the full dataset prior to the split. Zero-variance columns are removed before model fitting. The crime-enhanced feature set adds all five buffer count variables to the baseline.

The feature engineering approach serves two goals. The first is to represent listing characteristics in a form suitable for both regression and machine learning models. The second is to construct a feature matrix that can be tested with and without the crime variables in order to evaluate their contribution to prediction performance. The baseline feature set represents what is already available from platform data and standard listing attributes. The crime feature set represents the additional spatial information that requires external data integration and geographic processing. Comparing the two directly in the same model framework is what allows the study to isolate the contribution of crime.

3.4 Modeling Strategy

3.4.1 Regression Benchmarks

Three nested linear regression models are estimated. The baseline specification includes crime exposure at the 500m buffer along with accommodates and bedrooms. The controls model adds the full set of structural and platform variables (bathrooms, beds, room type, review score, number of reviews, availability). The neighborhood fixed-effects model adds indicator variables for each neighborhood. Additional buffer comparison models replace the primary crime variable with each of the alternative buffer measures (100m, 200m, 300m, 1,000m) under the same control structure, each fitted as a separate regression to avoid collinearity among the nested buffers. All regression models use log price as the dependent variable. These benchmarks establish the statistical significance and direction of the crime-price relationship across buffer specifications and provide a transparent baseline for comparison with the machine learning models.

3.4.2 Gradient Boosting Models

The primary predictive models are gradient boosting machines implemented using the XGBoost library (Chen & Guestrin, 2016), an efficient and regularized implementation of the gradient boosting framework introduced by Friedman (2001). Three GBM specifications are estimated and compared. The baseline GBM includes only the standard listing and neighborhood features. The crime-enhanced GBM adds all five violent-crime exposure buffer counts. The PCA-based GBM applies principal component analysis to the crime-enhanced feature matrix before model fitting, using components that explain 90 percent of cumulative variance as the feature set. PCA is fitted on the training data only and applied to the test data to prevent leakage. The comparison between the first and second models directly tests whether crime improves prediction. The comparison with the PCA model tests whether dimensionality reduction is a useful alternative to the original feature engineering approach.

The data are split 80/20 into training and test subsets using a fixed random seed for reproducibility. The split yields 3,683 listings in the training set and 921 listings in the test set. All XGBoost models use the regression objective reg:squarederror and the gbtree booster.

3.4.3 Principal Component Analysis

The third gradient boosting specification uses principal component analysis (PCA) as a dimensionality-reduction alternative to the engineered feature set. Rather than feeding the model the original structural, neighborhood, and crime variables, PCA re-expresses the crime-enhanced feature matrix as a set of orthogonal components ordered by the share of total feature variance each explains, and the model is then trained on the leading components that cumulatively account for 90 percent of variance. The motivation for including this specification is twofold. First, the crime buffers are nested and therefore collinear by construction, and PCA is a standard remedy for collinearity that re-projects correlated inputs onto uncorrelated axes. Second, comparing the engineered features against a compressed representation tests whether the manual feature engineering is actually doing useful work.

Two procedural points are important. PCA is fitted on the training data only and the resulting rotation is then applied to the test data, so that no test-set variance informs the components and no information leaks across the split. The components are computed on centered and scaled inputs, because PCA is sensitive to the relative scale of the variables. A practical consequence of this design, explored in the results, is that PCA only helps when the feature space is genuinely low-dimensional; if variance is spread across many components rather than concentrated in a few, the components discarded to achieve compression still carry predictive signal, and PCA can underperform the original features.

3.4.4 Hyperparameter Tuning

Hyperparameters are selected through a grid search across 48 combinations of five parameters: maximum tree depth (3, 5, 6), learning rate (0.05, 0.10), row subsample fraction (0.70, 0.85), column subsample fraction (0.70, 0.85), and minimum child weight (1, 5). The number of trees is determined separately by early stopping rather than included in the grid. For each combination, 5-fold cross-validation with early stopping (30 rounds, maximum 500 trees) is applied to the training portion of the crime-enhanced model. The combination minimizing mean CV RMSE is selected and applied identically to all three GBM specifications. This shared-hyperparameter approach trades off a small potential advantage to the crime-enhanced model (since tuning was done on its feature space) against the clearer comparison it enables. Any performance differences across models can be attributed to feature content rather than to tuning differences.

3.4.5 Spatial Block Cross-Validation

Standard random cross-validation ignores spatial autocorrelation. Listings close together share similar crime buffer counts by construction, so a random train-test split places training observations spatially adjacent to many test observations, leaking spatial information and potentially inflating the predictive value of crime features. Spatial block cross-validation addresses this by overlaying a regular grid of 5 km by 5 km blocks on the listing locations and randomly assigning each block to one of five folds, using the blockCV package in R (Valavi et al., 2019). Each fold is held out in turn as the test set while the model is trained on the remaining four folds. This forces the model to generalize across spatially separated blocks rather than interpolate within them. The 5 km block size is intended to be larger than the spatial autocorrelation range of the crime features while still producing folds of adequate sample size. This evaluation directly addresses whether any improvement from crime features reflects genuine predictive signal or an artifact of spatial proximity in random splitting.

3.4.6 Random Forest Comparison

To establish that the central spatial finding is not specific to one learning algorithm, the baseline and crime-enhanced specifications are additionally estimated with random forest, a structurally distinct tree-based ensemble implemented through the ranger package in R (Breiman, 2001). Whereas gradient boosting builds trees sequentially, with each tree correcting the residuals of the ensemble so far, random forest grows many deep trees independently on bootstrap resamples of the training data and averages their predictions. Because the two methods construct and combine trees in fundamentally different ways, agreement between them on the spatial result provides stronger evidence that the result reflects the structure of the data rather than the idiosyncrasies of a single estimator.

The comparison is held as controlled as possible. Both algorithms use the identical feature matrices, the identical 80/20 random split, and the identical 5 km spatial blocks. The random forest grows 500 trees with a minimum node size of five. Its one principal tuning parameter, the number of candidate variables considered at each split (`mtry`), is selected by minimizing out-of-bag root mean squared error on the crime-enhanced training set, with the chosen value then applied to both the baseline and crime-enhanced random forests. This mirrors the shared-tuning logic used for the gradient boosting models: out-of-bag error provides an internal validation estimate that random forest produces without a separate cross-validation loop, so no additional data partitioning is required for tuning. Each random forest specification is evaluated under both the random split and spatial block cross-validation, using the same R^2, RMSE, and MAE metrics applied to the gradient boosting models, so that the crime contribution can be compared across algorithms and across evaluation schemes on a common footing.

3.4.7 SHAP Values

Feature importance is quantified using SHAP (SHapley Additive exPlanations) values computed via the TreeSHAP algorithm (Lundberg & Lee, 2017; Lundberg, Erion, & Lee, 2018). SHAP values decompose each prediction into additive per-feature contributions, providing both global importance (mean absolute SHAP) and directional information (signed mean SHAP). This is preferable to XGBoost's built-in gain-based importance, which is known to be biased toward continuous and high-cardinality variables. Dependence plots are produced for each crime variable to visualize how each variable's contribution varies with its value, with point coloring indicating a potentially interacting feature. One caveat is worth noting: SHAP values can be unstable when input features are highly correlated, which is relevant here because the five crime buffers are nested and therefore correlated by construction. SHAP rankings among the crime variables should therefore be interpreted as indicative rather than definitive.

3.4.8 Evaluation Metrics

Model performance is evaluated on the held-out test set using three complementary metrics: R^2, root mean squared error (RMSE), and mean absolute error (MAE). R^2 measures the proportion of variance in log nightly price explained by the model on the test set. RMSE penalizes large prediction errors more heavily than small ones, making it sensitive to outliers in log-price space. MAE measures average absolute prediction error and is less sensitive to outliers than RMSE. Using all three metrics together allows the thesis to distinguish between improvements in average error performance and improvements in the handling of extreme cases. If the crime-enhanced model produces improvements across all three metrics, that provides more credible and multidimensional evidence than any single statistic alone. The thesis does not separately compute formal significance tests on the difference in R^2 between models; the spatial block cross-validation analysis (Section 3.4.5) serves as a more demanding robustness check.

Chapter 4:

Results

This chapter presents the empirical results of the Dallas residential short-term rental price prediction analysis. The results are organized into six sections. Section 4.1 provides a descriptive overview of the analytical dataset and the spatial distribution of listings and violent crime across Dallas. Section 4.2 presents the regression benchmark results and buffer-zone comparison. Section 4.3 compares the three gradient boosting model specifications on the held-out test set. Section 4.4 presents the SHAP feature importance analysis and discusses what the crime variable rankings mean for interpreting the model. Section 4.5 presents the spatial block cross-validation results and explains their implications for evaluating crime-prediction models. Section 4.6 extends the analysis with a random forest comparison to establish that the central spatial finding is not an artifact of a single learning algorithm.

4.1 Descriptive Overview

The analytical dataset combines 5,290 residential short-term rental listings (in the regression sample) with Dallas violent-crime data. The median nightly listing price is $127, with prices ranging from $12 to the 99th-percentile cap of $1,581. Log price has a mean of 4.958. Bedroom counts range from studio configurations to large multi-bedroom properties, with the majority of listings in the one-to-three bedroom range. Room type is dominated by entire unit listings. Crime buffer counts show substantial variation across listings and distances, as detailed in Table 4.1.

Table 4.1: Descriptive Statistics — Key Variables

Variable	Mean	Median	SD	Min	Max
Nightly price (USD)	198.4	127.0	224.0	12.0	1,581.0
Log price	4.958	4.844	0.749	2.485	7.366
Bedrooms	1.89	1.00	1.34	0.00	33.0
Bathrooms	1.63	1.00	0.98	0.00	18.0
Review score	4.754	4.870	0.445	1.00	5.00
Crime count 100m	5.3	1.0	11.1	0.0	122.0
Crime count 500m	111.9	59.0	147.8	0.0	736.0
Crime count 1000m	345.5	248.0	348.7	0.0	1,505.0

The crime variables show considerable variation across listings, which is essential for the predictive analysis. The 100-meter counts are the most tightly distributed because fewer incidents fall within a very small radius of any given listing. As the buffer expands, the mean count increases and the distribution widens because more incidents are captured within the larger area. The 1,000-meter buffer produces the highest mean counts and the widest spread. This variation across both listings and buffer distances is what allows the model to learn the relationship between local crime exposure and price. If all listings had similar crime counts regardless of location, crime would not be a useful predictive feature at all.

The spatial distribution of listings and violent crime across Dallas reinforces the need for localized measurement rather than district-level averages. Residential short-term rental listings are concentrated in neighborhoods with stronger demand, better accessibility, and higher market activity. Violent crime is also spatially clustered rather than evenly dispersed across the city. Some areas experience much higher local exposure than others. A citywide average crime rate would be analytically weak in this context because it would treat listings in very different local environments as if they faced identical safety conditions. The listing-centered buffer approach preserves the within-neighborhood variation that aggregate measures would conceal.

The modeling samples used in the analysis differ between the regression benchmarks and the gradient boosting models. The regression benchmarks use 5,290 listings, with implicit listwise deletion across nested specifications. The GBM sample is restricted further to 4,604 listings because of additional listwise deletion on review-related variables, with 3,683 in the training set and 921 in the held-out test set after the 80/20 split. The variation in crime buffer counts across these listings spans a wide range, from listings with zero incidents in even the broadest buffer to listings with more than 1,500 incidents within 1,000 meters. This range of exposure is essential for the prediction model to learn a meaningful relationship between crime and price rather than treating all listings as similarly situated in terms of neighborhood safety.

4.2 Regression Benchmark Results

The regression benchmarks establish that violent crime is negatively and significantly associated with listing price across all buffer specifications, and they confirm that the data behave in a theoretically consistent way before machine learning models are applied. The baseline regression explains 41.5 percent of the variation in log price. This means that a very simple model with only a few listing attributes captures a meaningful share of price variation but leaves most of it unexplained. Once standard listing controls are added, the

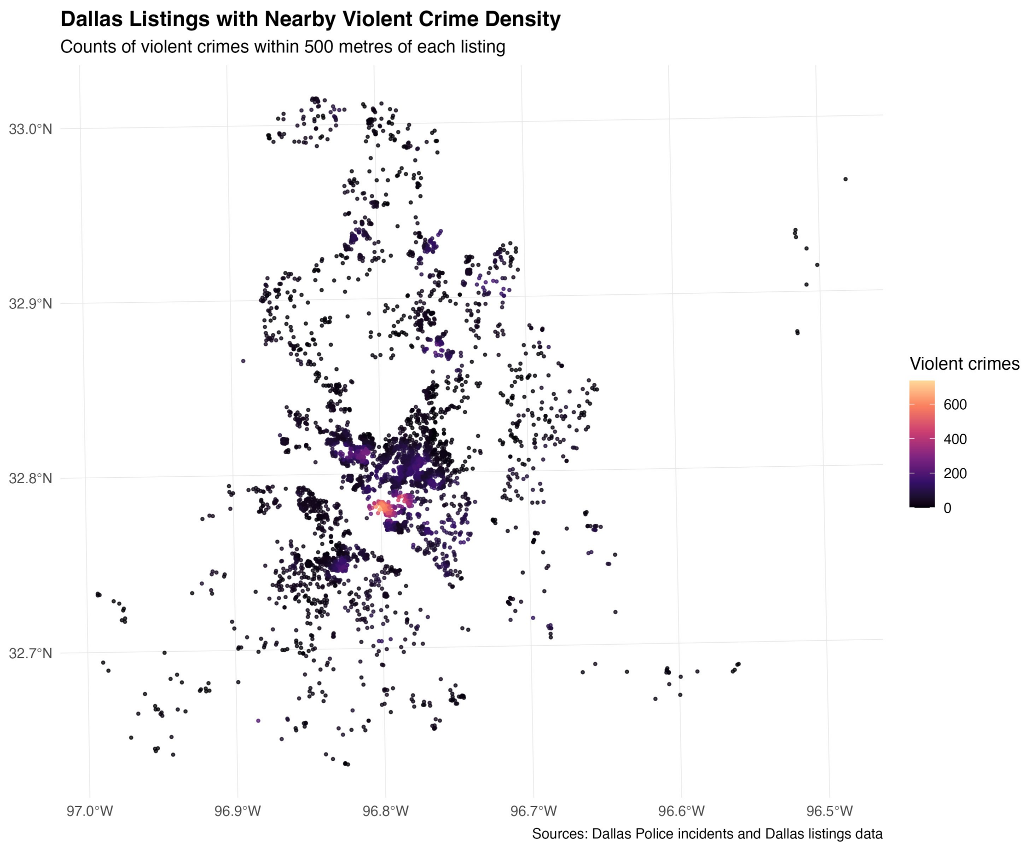

Figure 4.1: Spatial distribution of Dallas Airbnb listings coloured by violent-crime count within 500 meters. High-crime concentrations are visible in the central city corridor.

R^2 increases to 58.2 percent. This large improvement confirms that structural features like bedrooms, bathrooms, capacity, review scores, and availability are very important to short-term rental pricing and carry a substantial amount of predictive information on their own.

Table 4.2: Regression Benchmark Model Comparison

Model	Obs.	R^2	Adj. R^2	Resid. SE
Baseline	5,290	0.415	0.415	0.573
Controls	5,290	0.582	0.581	0.467
Neighborhood Fixed Effects	5,290	0.609	0.607	0.452
Buffer 100m (FE)	5,290	0.607	0.605	0.453
Buffer 200m (FE)	5,290	0.608	0.606	0.453
Buffer 300m (FE)	5,290	0.609	0.607	0.452
Buffer 500m (FE)	5,290	0.609	0.607	0.452
Buffer 1000m (FE)	5,290	0.608	0.606	0.452

Adding neighborhood fixed effects raises R^2 further to 60.9 percent, indicating that spatial location captures information about price that goes beyond what listing-level variables alone can explain. This is consistent with hedonic pricing theory, where price is shaped by the surrounding environment and location. The addition of neighborhood fixed effects essentially absorbs much of the broad spatial variation in the data, which is exactly what creates the challenge for crime features. After neighborhood effects are accounted for, crime must explain within-neighborhood variation in price, which is a harder standard to meet than explaining variation across the full city.

All five crime coefficients are negative and statistically significant across every buffer specification, as shown in Table 4.3. This is an important confirmation that the data behave in a theoretically consistent way. Higher violent-crime exposure is associated with lower listing prices, and this relationship is not sensitive to which spatial scale of crime measurement is used. The 100-meter buffer has the largest negative coefficient (-0.00263), indicating the strongest per-incident association with price at the most immediate spatial scale. This makes sense because a listing that is extremely close to a violent incident is more directly exposed to the safety risk than one where the incident occurred several blocks away. As the buffer expands, the coefficient magnitude decreases, reflecting dilution of the per-incident effect over larger areas.

The 500-meter buffer produces the highest model fit among the comparison models, suggesting that broader neighborhood context is also informative for price. Still, the differences in R^2 across buffer distances are small. This means crime matters at multiple spatial scales rather than only at one. The regression benchmarks are useful for several reasons. First, they show that the data behave in a theoretically reasonable way. Second, they confirm that

Table 4.3: Buffer Zone Regression Results

Buffer	Crime Variable	Coeff.	Std. Err.	t-stat	p-value	CI Lower	CI Upper
100 m	crime_density_100m	−0.00263	0.00070	−3.73	< 0.001	−0.00400	−0.00125
200 m	crime_density_200m	−0.00109	0.00023	−4.78	< 0.001	−0.00153	−0.00064
300 m	crime_density_300m	−0.00069	0.00012	−5.70	< 0.001	−0.00093	−0.00046
500 m	crime_density	−0.00036	0.00006	−6.26	< 0.001	−0.00047	−0.00025
1000 m	crime_density_1000m	−0.00008	0.00002	−4.61	< 0.001	−0.00012	−0.00005

Note: All models include the same controls and neighborhood fixed effects; only the crime buffer distance changes. Dependent variable is log nightly price.

violent crime has a negative and significant relationship with price across all buffer specifications. Third, they establish a baseline that can be compared with the machine learning models.

The pattern of decreasing coefficient magnitude across buffer distances also has a straightforward interpretation. A listing surrounded by a small number of incidents at very close range is more directly exposed to safety risk than one where the same incidents fall within a much larger area. As the buffer radius grows, each additional incident contributes proportionally less to the perceived safety environment of the listing, because more of the counted incidents are farther away and less directly associated with the immediate listing experience. This is consistent with how urban geography research conceptualizes the decay of spatial externalities with distance, and it provides reassurance that the crime variables are behaving in a way that reflects a real underlying mechanism rather than a statistical artifact.

4.3 Machine Learning Model Results

4.3.1 Hyperparameter Selection

The grid search across 48 combinations identifies the optimal GBM configuration through 5-fold cross-validation on the training portion of the crime-enhanced dataset. The best combination achieves a mean CV RMSE of 0.394. The selected configuration uses a maximum tree depth of 6, a learning rate of 0.05, a row subsample fraction of 0.70, a column subsample fraction of 0.85, a minimum child weight of 1, and 247 trees selected by early stopping. This configuration is applied identically to all three GBM specifications to ensure that any differences in performance reflect feature content rather than tuning differences.

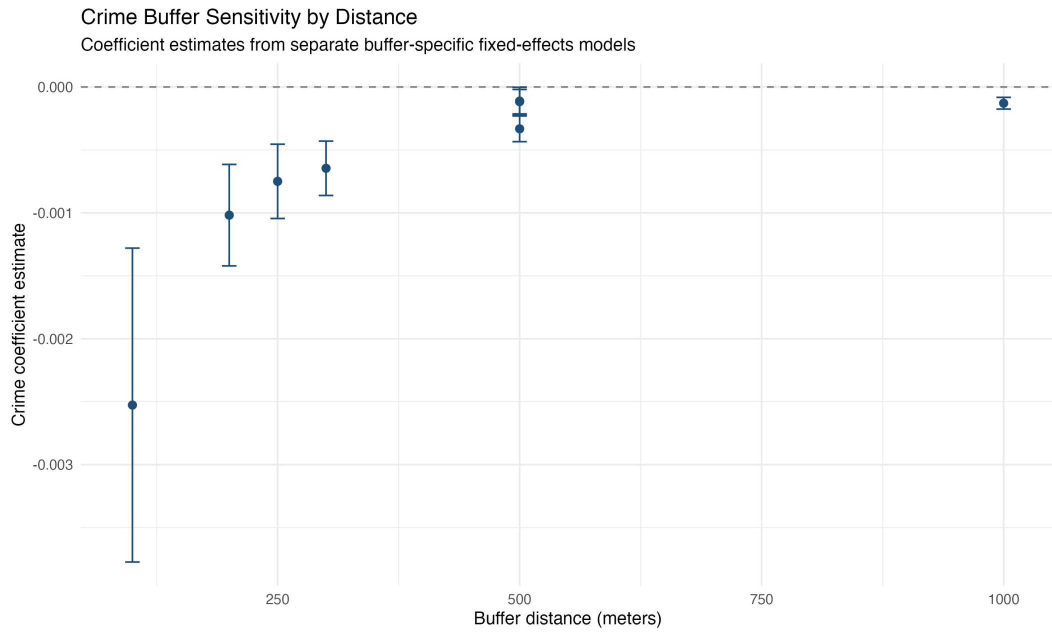

Figure 4.2: Crime coefficient estimates and 95% confidence intervals across buffer distances. All coefficients are negative and statistically significant; effect magnitude decreases with buffer distance.

4.3.2 Test Set Performance

The three GBM models are evaluated on the held-out test set of 921 listings. The results are presented in Table 4.4.

Table 4.4: Predictive Performance of GBM Models — Test Set

Model	R^2	RMSE	MAE
GBM Baseline	0.689	0.402	0.285
GBM + Crime Features	0.701	0.394	0.272
GBM + PCA Features	0.684	0.405	0.286

Note: Performance metrics calculated on held-out test set ($N = 921$). RMSE and MAE are in log-price units.

The crime-enhanced model achieves the highest R^2 (0.701), lowest RMSE (0.394), and lowest MAE (0.272) on the test set, outperforming the baseline on all three metrics. The improvement is modest rather than large. Adding crime features raises explained variance by 1.2 percentage points and reduces prediction error. This pattern is reasonable given that structural features already explain most of the pricing variation. The result shows that violent-crime exposure contains additional predictive information that the baseline model cannot fully capture on its own under random-split evaluation.

The PCA model performs worse than both the baseline and the crime-enhanced model, with R^2 of 0.684, RMSE of 0.405, and MAE of 0.286. This indicates that PCA did not improve prediction in this dataset. A principal component may summarize variation in the data without preserving the particular structure that matters for predicting price. In this case, 18 components were required to explain 90 percent of feature variance, which indicates that the feature space is genuinely multidimensional and does not compress cleanly into a small number of summary dimensions. The original engineered features are more useful than any compact approximation. Methodologically, this shows that the model was tested against a principled alternative rather than simply accepted by default.

The GBM models substantially outperform the regression benchmarks. The best regression model achieves an R^2 of 60.9 percent, while the best GBM model achieves 70.1 percent on the test set. This gap reflects the ability of gradient boosting to capture non-linear relationships and interactions that additive linear models cannot accommodate. The pricing structure of residential short-term rentals contains complexity that goes beyond what standard regression can represent, and the gradient boosting framework is better equipped to learn that structure from the data.

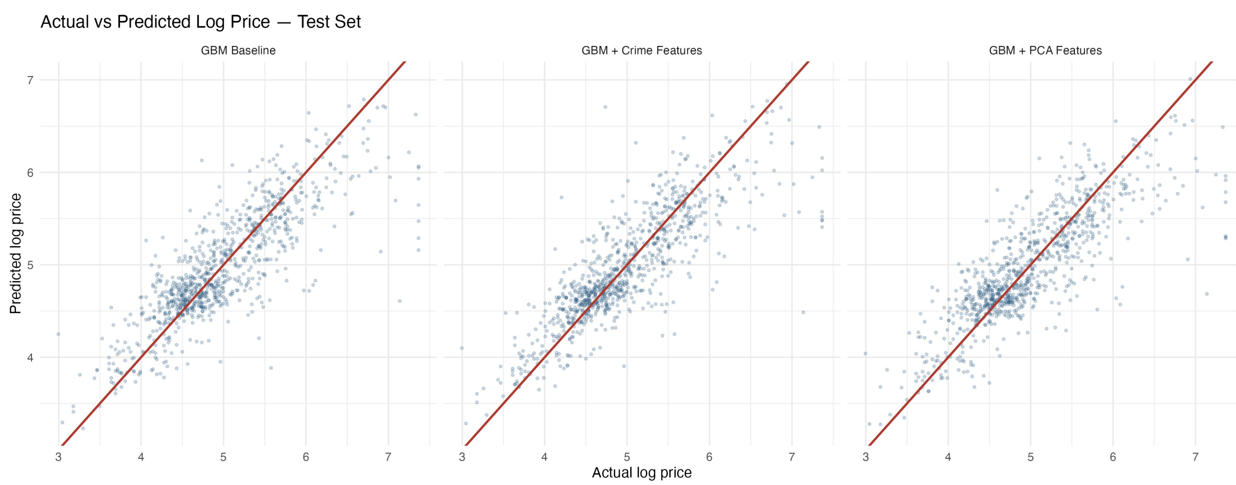

Figure 4.3: Actual versus predicted log price on the test set for all three GBM specifications. Points close to the 45-degree line indicate accurate predictions. The crime-enhanced model shows modestly tighter clustering than the baseline.

4.4 SHAP Feature Importance

SHAP values are computed via the TreeSHAP algorithm on the training set of the crime-enhanced GBM. The results confirm that structural characteristics dominate the pricing signal. Bedrooms is the most important predictor, with a mean absolute SHAP value of 0.242, more than twice that of the next predictor, bathrooms (0.107). These results make

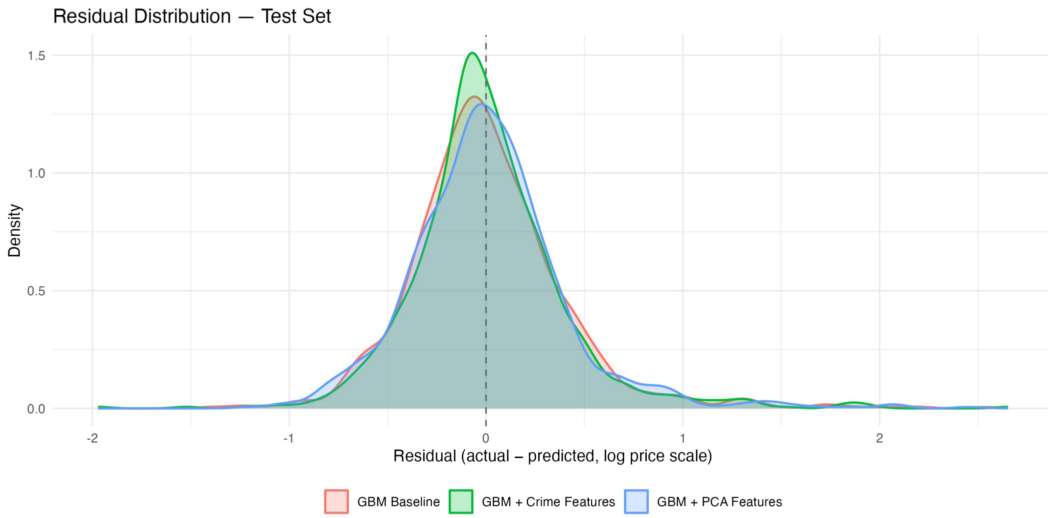

Figure 4.4: Residual distribution (actual minus predicted) on the test set for all three GBM specifications. All three distributions are centered near zero; the crime-enhanced model shows a marginally tighter peak.

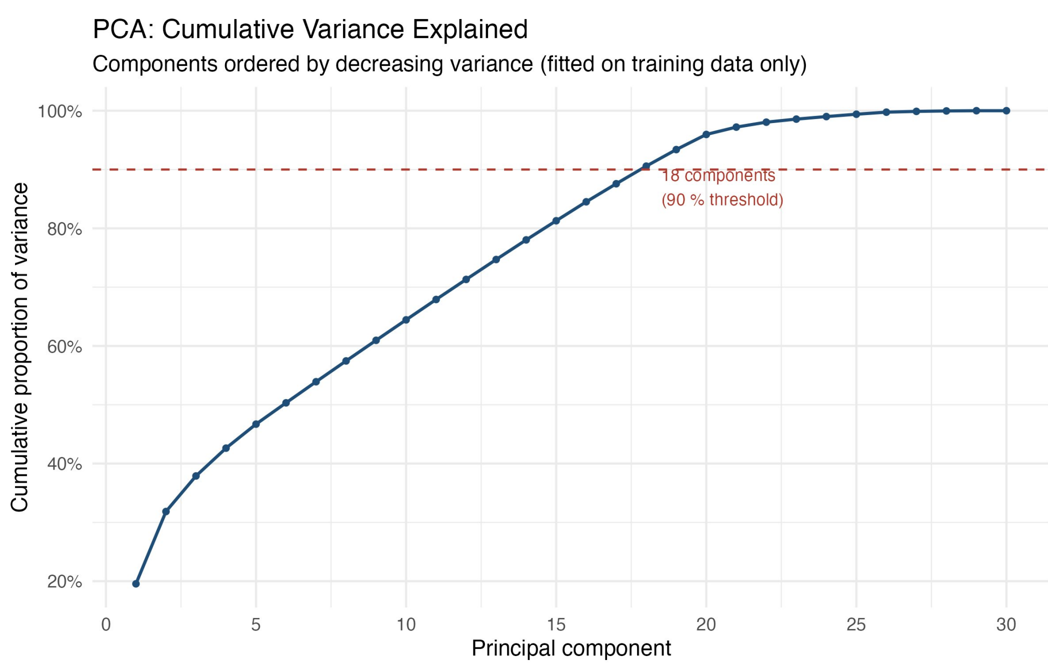

Figure 4.5: Cumulative variance explained by principal components, fitted on training data only. Eighteen components are required to explain 90 percent of variance, indicating that the feature space is multidimensional.

sense because bedroom count is one of the strongest indicators of the size and capacity of a residential short-term rental listing. Guests searching for accommodation have specific size requirements, and bedroom count is typically the primary filter they use to identify suitable listings.

Table 4.5: SHAP Feature Importance Rankings — Crime-Enhanced GBM (Top 15)

Rank	Feature	Mean |SHAP|	Mean SHAP
1	bedrooms	0.2420	+0.2420
2	bathrooms_num	0.1073	+0.1073
3	room_type: Entire home/apt	0.0844	varies
4	review_scores_rating	0.0822	+0.0822
5	accommodates	0.0779	+0.0779
6	neighborhood: District 14	0.0612	+0.0612
7	availability_365	0.0374	varies
8	crime_density_1000m	0.0347	+0.0027
9	crime_density (500 m)	0.0289	+0.0060
10	neighborhood: District 2	0.0287	varies
11	crime_density_100m	0.0238	−0.0041
12	crime_density_300m	0.0233	+0.0022
13	beds	0.0220	varies
14	number_of_reviews	0.0220	varies
15	crime_density_200m	0.0220	+0.0048

Note: Mean SHAP sign indicates average directional effect. Values computed on training set ($N = 3,683$).

Crime variables rank 8th, 9th, 11th, 12th, and 15th out of all features in the model. This means that even though crime features are not the primary drivers of prediction, they still play a meaningful role as secondary predictors. The 1,000-meter buffer ranks highest among crime variables (mean |SHAP| = 0.035), followed by the 500-meter buffer (0.029). The combined relative contribution of all crime variables places them collectively on par with bathrooms alone. This confirms the interpretation that crime is a meaningful secondary predictor that contributes useful information at the margin in the random-split setting.

The direction of crime effects across buffer sizes shows a more nuanced pattern than the regression benchmarks alone would suggest. Only the 100-meter buffer (mean SHAP = −0.0041) contributes negatively to predicted price on average, consistent with the intuition that immediate local crime deters demand and depresses prices. A violent incident occurring directly outside a listing plausibly affects perceived safety more than one occurring several blocks away. The remaining buffers, ranging from 200m to 1,000m, show positive average SHAP values, which seems counterintuitive at first. SHAP dependence plots reveal that this reflects a confound with urban centrality. Dallas's most active, expensive, and high-demand areas, including District 14 (Uptown) and surrounding neighborhoods, also have moderate-to-high crime counts at broader radii simply because they are densely populated urban centers. The model partially learns to associate broad crime exposure with central location and therefore higher prices for entire-home listings. The negative regression coefficients on the same buffers, by contrast, are conditional on neighborhood fixed effects, which absorb much of this centrality confound. The contrast between the unconditional SHAP attribution and the conditional regression coefficient is itself informative since it shows that the crime-price relationship is genuinely negative within neighborhoods but is masked, at broader spatial scales, by between-neighborhood differences in centrality and density.

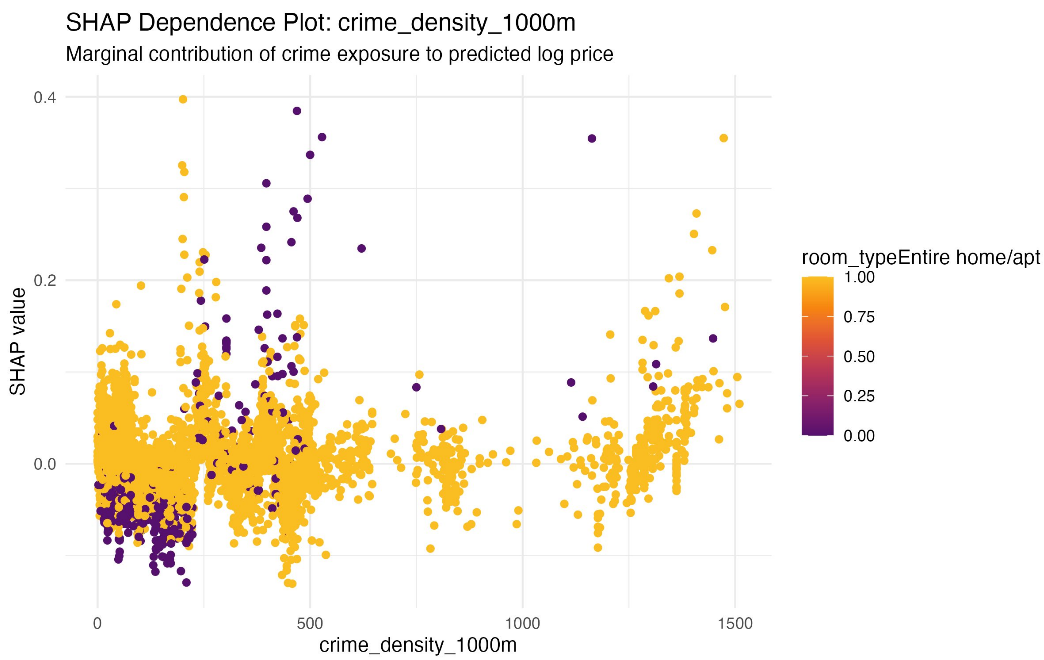

Figure 4.6: SHAP dependence plot for the 1,000 m crime buffer, colored by room type. Entire-home listings show positive SHAP values across a wide range of crime counts, reflecting the confound between broad crime exposure and urban centrality in Dallas.

4.5 Spatial Block Cross-Validation

Spatial block cross-validation divides Dallas into 5 km grid blocks using the blockCV package, with blocks randomly assigned to one of five folds (Valavi et al., 2019). Each fold is used as a held-out test set in turn, forcing the model to generalize across spatially separated blocks rather than interpolate within them. This is the most demanding evaluation in the study, and the results are the most important finding of the thesis.

The spatial block cross-validation results show that the crime-enhanced model and the

Table 4.6: Spatial Block CV Summary (5-Fold, 5 km Blocks)

Model	R^2 Mean	R^2 SD	RMSE Mean	RMSE SD	MAE Mean	MAE SD
GBM Baseline	0.574	0.127	0.468	0.093	0.331	0.034
GBM + Crime Features	0.574	0.148	0.468	0.108	0.328	0.041

Note: Five folds created by random assignment of 5 km grid blocks using the blockCV package. Metrics averaged across folds; SD reflects fold-to-fold variation.

baseline model achieve identical mean spatial CV R^2 (0.574), RMSE (0.468), and a slight improvement in MAE (0.331 to 0.328). The improvement observed under random test-set evaluation from 0.689 to 0.701 disappears entirely when geographic blocks are used to create the holdout. This is the most important finding in the thesis, and it substantially qualifies the random-split results. It indicates that the crime contribution to prediction under random splitting is largely an artifact of spatial autocorrelation. Because listings close together in space share similar crime buffer counts and similar prices, a random split allows the model to exploit proximity rather than genuinely learning a crime-price relationship that generalizes across the city.

The crime-enhanced model also shows higher fold-to-fold variance (R^2 SD = 0.148 vs. 0.127 for the baseline), indicating less consistent performance across geographic areas when the spatial leakage is removed. The crime-enhanced model is not only failing to improve over the baseline under spatial holdout; it is also more unstable across folds. This finding does not mean crime is economically irrelevant. The regression benchmarks consistently show negative and significant crime coefficients across all five buffer distances. These hedonic results confirm that violent crime functions as a neighborhood disamenity in Dallas residential short-term rental markets. The distinction is between explanatory significance and predictive generalizability across geographic space.

4.6 Algorithm Comparison: Random Forest

The preceding analysis evaluates the contribution of localized violent-crime features using gradient boosting. Because the central methodological claim of this thesis concerns whether the apparent predictive value of spatially correlated features survives spatial evaluation, it is important to establish that the finding is not an artifact of one particular learning algorithm. This section therefore re-estimates the baseline and crime-enhanced models using random forest, a structurally distinct tree-based ensemble, and compares the two algorithms under both random and spatial evaluation.

Random forest builds an ensemble of independently grown deep trees through bootstrap

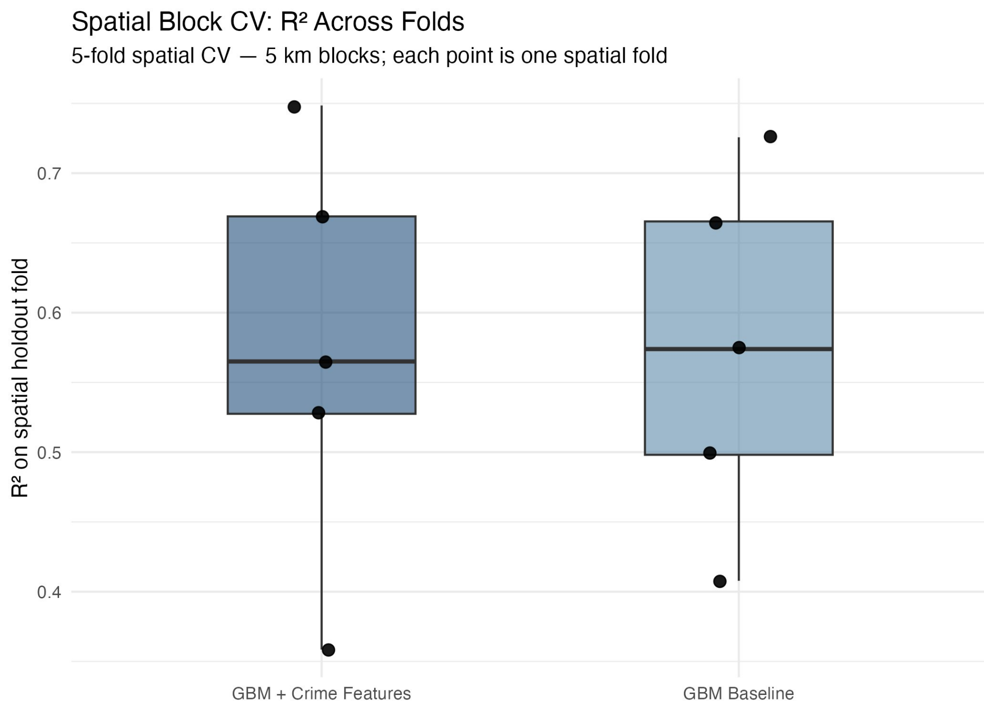

Figure 4.7: Spatial block CV R^2 across five folds for the GBM Baseline and GBM + Crime Features models. The distributions are nearly identical in median and overlap substantially, confirming that the crime contribution does not survive spatial holdout.

aggregation, in contrast to the sequential, gradient-based construction used by gradient boosting. The two methods therefore exploit the feature space in different ways, which makes random forest a useful check on the generality of the spatial finding. To keep the comparison controlled, both algorithms use the identical feature matrices, the identical 80/20 random split, and the identical 5 km spatial blocks described in Chapter 3. The random forest uses 500 trees with a minimum node size of five; the number of variables considered at each split (*mtry*) was selected by minimizing out-of-bag root mean squared error on the crime-enhanced training set, mirroring the shared-tuning logic applied to the gradient boosting models.

Table 4.7 reports test-set R^2 for both algorithms under each evaluation scheme. Under the conventional random split, the addition of crime features improves predictive accuracy for both learners, and the improvement is in fact larger for random forest (a gain of 0.032 in R^2, from 0.675 to 0.707) than for gradient boosting (a gain of 0.012, from 0.689 to 0.701). Considered in isolation, this random-split evidence would suggest that crime features are a robust and even substantial predictive asset.

Under spatial block cross-validation, however, the improvement vanishes for both algorithms. Table 4.8 isolates the crime contribution—the difference in R^2 between the crime-enhanced and baseline specifications—under each evaluation scheme. For gradient boosting,

Table 4.7: Predictive Performance Across Algorithms and Evaluation Schemes

Algorithm	Feature set	Random-split R^2	Spatial-CV R^2
Gradient Boosting	Baseline	0.689	0.574
Gradient Boosting	+ Crime	0.701	0.574
Gradient Boosting	PCA	0.684	—
Random Forest	Baseline	0.675	0.582
Random Forest	+ Crime	0.707	0.578

Note: Random-split values are test-set R^2 ($N = 921$). Spatial-CV values are the mean across five 5 km spatial-block folds. The PCA specification has no spatial-CV entry because it was estimated only under the random split. Both algorithms use identical features, the same random split, and the same spatial folds.

the crime gain falls from +0.012 under random splitting to −0.001 under spatial holdout; for random forest, it falls from +0.032 to −0.004. In both cases the crime contribution is statistically indistinguishable from zero once geographic separation is enforced, and in both cases its point estimate is marginally negative.

Table 4.8: Crime Contribution by Algorithm and Evaluation Scheme

Algorithm	Evaluation	Baseline R^2	+ Crime R^2	Crime gain (ΔR^2)
Gradient Boosting	Random split	0.689	0.701	+0.0116
Gradient Boosting	Spatial CV	0.574	0.574	−0.0005
Random Forest	Random split	0.675	0.707	+0.0319
Random Forest	Spatial CV	0.582	0.578	−0.0039

Note: Crime gain is the difference in R^2 between the crime-enhanced and baseline specifications. Positive values indicate that crime features improve prediction. Under spatial holdout the gain is negligible and marginally negative for both algorithms.

Figure 4.8 displays the same pattern: the two algorithms diverge under random splitting—random forest showing the larger crime gain—yet converge to an essentially identical spatial-CV plateau near 0.58, with the crime and baseline bars indistinguishable within each algorithm.

This convergence is the substantive result of the comparison. The magnitude of the apparent crime gain under random splitting is algorithm-dependent: random forest, which relies more heavily on spatial proximity in this setting, exhibits a larger illusory improvement. The disappearance of that gain under spatial evaluation, by contrast, is not algorithm-dependent. Both learners reach the same spatial plateau, and for both the crime contribution is erased. This indicates that the inflation of spatially correlated features under random train–test splitting is a property of the data structure rather than of any single estimator, and it reinforces the central methodological recommendation of this thesis: spatial cross-

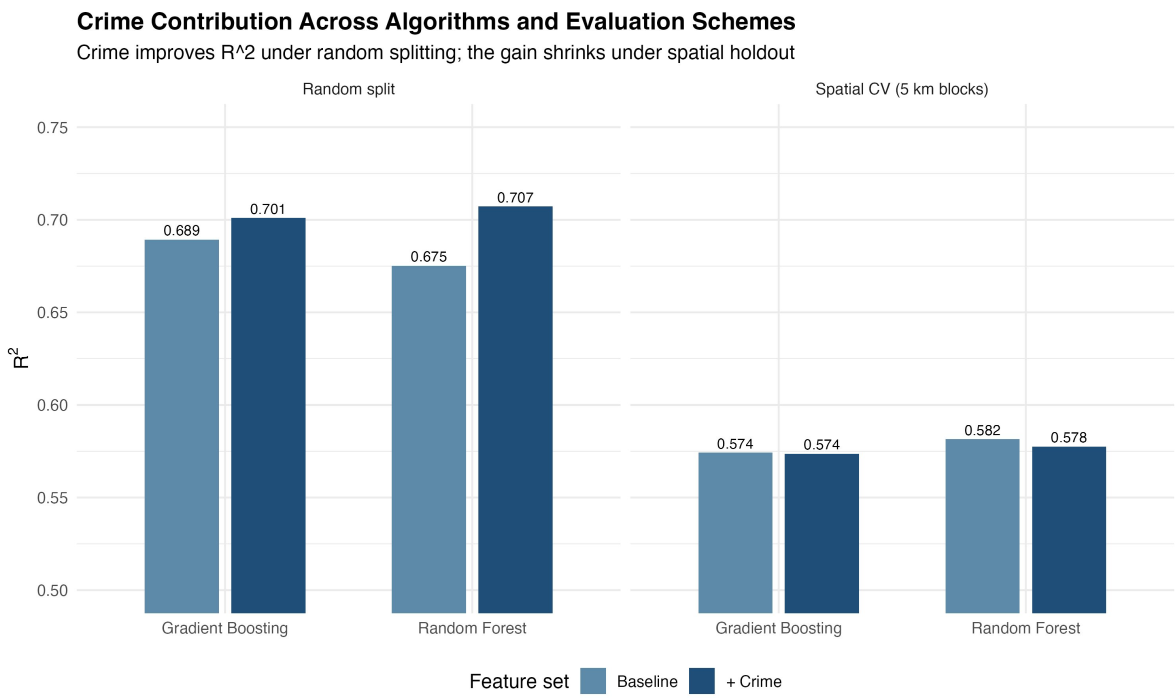

Figure 4.8: Test-set R^2 for gradient boosting and random forest, with and without crime features, under random splitting (left) and spatial block cross-validation (right). Crime features improve R^2 for both algorithms under random splitting—more so for random forest—but the gain disappears under spatial holdout, where the two algorithms converge to a common plateau and the crime and baseline bars become indistinguishable.

validation should be treated as a standard robustness check whenever spatially correlated features are evaluated in a predictive model, irrespective of the learning algorithm employed.

Chapter 5:

Discussion

5.1 Interpretation of Main Findings

The central question of this thesis, whether localized violent-crime exposure improves the prediction of residential short-term rental prices in Dallas, has a nuanced answer. Under random test-set evaluation, crime features improve prediction consistently across R^2, RMSE, and MAE. Neighborhood safety is a meaningful characteristic that contributes to the bundle of attributes determining listing price. The gradient boosting model captures this relationship as part of a broader nonlinear pricing system in which structural features remain dominant. The contribution of crime is real enough to improve performance on three separate metrics, which makes it more credible than a result that appears in only one.

However, the spatial block cross-validation finding qualifies this result substantially. When geographic blocks are used to create the holdout, the crime improvement vanishes. Both the baseline and crime-enhanced models achieve a mean spatial CV R^2 of 0.574, with overlapping distributions across folds. This reveals that the improvement observed under random splitting is largely an artifact of spatial autocorrelation. Dallas listings close in space share similar crime counts and similar prices, so a random split allows the model to exploit proximity rather than learning a crime-price relationship that generalizes across the city. This is the most important finding in the thesis. It does not invalidate the regression results, which use a different methodology and are not subject to spatial leakage in the same way. But the claim that crime improves a machine learning prediction model needs to be qualified carefully.

This finding does not mean crime is economically irrelevant. The regression benchmarks consistently show negative, significant crime coefficients across all five buffer distances, with effects that are correctly signed and scaled. These results provide credible evidence that violent crime functions as a neighborhood disamenity in Dallas residential short-term rental markets. The distinction is that crime matters for explaining prices at any given location, but its contribution to a machine learning model's cross-spatial generalization is limited by the data structure itself.

5.2 SHAP and the Direction of Crime Effects

The SHAP analysis adds another layer to the crime-price relationship. Only the 100-meter buffer shows a negative average SHAP value, which is consistent with the intuition that immediate local crime deters demand and depresses prices. A violent incident occurring directly outside a listing plausibly affects perceived safety more than one occurring several blocks away. This is also consistent with the regression results, where the 100-meter buffer produces the largest per-incident negative coefficient.

The remaining buffers at 200, 300, 500, and 1,000 meters show positive average SHAP values, which reflects a confound with urban centrality. Dallas's most active, expensive, and high-demand areas, including District 14 (Uptown) and surrounding neighborhoods, are concentrated in central parts of the city that also have moderate-to-high crime counts at broader radii simply because they are densely populated urban centers. The model partially learns to associate broad crime exposure with central location and therefore higher prices for entire-home listings. This is not a contradiction with the regression results, those coefficients are conditional on neighborhood fixed effects. The contrast illustrates that the crime-price relationship in Dallas is negative within neighborhoods but is masked at broader spatial scales by the spatial correlation between crime density and urban centrality. The implication for the predictive analysis is that broader buffers contribute less of a true crime signal and more of a centrality signal, which helps explain why the random-split improvement is so easily erased by spatial holdout.

5.3 Machine Learning Framework and PCA

The gradient boosting models substantially outperform regression benchmarks, with R^2 rising from 60.9 to 70.1 percent on the test set. This gap reflects the ability of gradient boosting to capture nonlinear relationships and interactions that additive linear models cannot accommodate. The pricing structure of residential short-term rentals contains complexity that goes beyond what standard regression can represent, and gradient boosting is better suited to learn that structure. The crime-enhanced GBM outperforms the baseline GBM on the random test set, confirming that crime adds predictive value within this evaluation framework even if that value disappears under spatial holdout.

The hyperparameter tuning process also deserves brief discussion. The grid search identifies a model configuration with a relatively low learning rate (0.05) and a modest tree depth (6), which suggests that the gradient boosting model is learning slowly and carefully from the data rather than fitting aggressively to noise. The 247 trees selected by early stopping indicates that the model has enough capacity to capture the pricing structure in the training data without overfitting. The fact that the same hyperparameter configuration is applied to all three GBM specifications means that any performance differences between the baseline, crime-enhanced, and PCA models can be attributed to feature content rather than tuning. This is an important design choice because it prevents the crime-enhanced model from benefiting from a hidden tuning advantage.

The PCA model performs worse than both the baseline and crime-enhanced models with R^2 of 0.684. Although 18 components are retained explaining 90 percent of variance, the cumulative variance plot reveals that variance accumulates gradually across components rather than concentrating in a few. This multidimensional structure means that PCA compression discards some predictive signal alongside noise. The result reinforces the methodological choice to use the original engineered features rather than a compressed representation. The PCA comparison was a legitimate robustness check rather than an expected competitor, and the result justifies the primary feature engineering approach.

The random forest comparison in Section 4.6 extends this framework to a structurally different tree-based learner and reaches the same conclusion. Random forest shows a larger crime improvement than gradient boosting under random splitting, yet that improvement is erased under spatial holdout in exactly the same way, with both algorithms converging to a common spatial-CV plateau. This indicates that the spatial-leakage finding is a property of the data rather than of any single estimator, strengthening the generality of the thesis's central methodological claim.

5.4 Limitations

Several limitations should be acknowledged. First, the study uses listed prices rather than realized transaction prices. Hosts may post prices that differ from what guests ultimately pay after discounts, cancellations, or platform adjustments. This means the model predicts advertised prices rather than final revenue. Second, the crime measure reflects reported violent incidents, which are subject to reporting variation across neighborhoods. Some neighborhoods may systematically underreport incidents, and some safety concerns relevant to guests may not appear in official police records at all. The violent-crime variables are strong proxies for neighborhood risk, but they are not perfect measures of actual danger or perceived risk. Third, the cross-sectional design cannot establish causal effects. The study tests whether crime improves prediction, not whether changes in crime cause changes in price. A panel design with calendar-level data could test the dynamic relationship between incident timing and price adjustments, but the available data do not support that approach.

Fourth, Euclidean buffers simplify the urban environment by ignoring roads, physical barriers, and how people actually navigate space. A 100-meter buffer drawn from a listing's coordinates may include areas that are physically inaccessible from the listing due to barriers, buildings, or transit infrastructure. Network-based distance measures that account for road geometry and pedestrian routes would provide a more realistic representation of proximity, but they require additional data and computational resources that go beyond the scope of this study. Listings near the Dallas city boundary may also have artificially low crime counts because buffers extend into areas where police incident data are not available. Fifth, the five nested crime buffers are correlated by construction (a 1,000m buffer fully contains a 100m buffer), which can destabilize SHAP attribution among them; SHAP rankings within the crime variable set should therefore be interpreted as indicative rather than definitive. Sixth, the results are specific to Dallas and may not generalize identically to other cities with different housing markets, tourism levels, regulatory environments, or crime distributions. Dallas is a strong setting for this research, but future work should test the same framework in additional cities to evaluate external validity.

The most significant methodological limitation is the spatial autocorrelation issue identified through spatial block cross-validation. Future work should routinely report both random and spatial cross-validation results when evaluating crime-prediction models, as random splits can substantially overstate the contribution of spatially correlated features. This has broader implications for the urban prediction literature, which frequently relies on random splits without testing for spatial generalizability.

5.5 Future Research

Future studies should seek access to higher-resolution data, including realized transaction prices rather than posted prices, cost-of-living data that could allow more precise centrality control, and panel data that could support longitudinal analysis of how listing prices respond to changes in local crime over time. Network-based distance measures that account for road geometry and pedestrian routes would improve on Euclidean buffers. Geographically weighted regression or spatial interaction models could formally test whether the crime-price relationship is spatially heterogeneous across Dallas neighborhoods.

The finding that broader crime buffers act as centrality proxies suggests a productive avenue for methodological refinement. Including explicit centrality measures such as distance to the central business district or transit access scores alongside crime buffers may allow the model to separate safety and location effects more cleanly. This would allow a sharper test of the crime-price hypothesis and produce more actionable results for hosts, platforms, and urban policymakers. It would also reduce the entanglement between crime counts and urban density that makes the interpretation of SHAP values for broader buffers difficult in the current analysis.

A further refinement would be to use non-overlapping ring features (0–100m, 100–200m, 200–300m, etc.) instead of nested cumulative buffers. Non-overlapping rings would reduce the collinearity among crime variables and produce more interpretable SHAP attributions for each spatial scale.

Beyond the Dallas case, the spatial cross-validation methodology demonstrated in this thesis could be applied to other urban prediction problems where spatially correlated features are included in machine learning models. Urban research increasingly uses features such as proximity to amenities, transit access, environmental exposure, and neighborhood demographics that are all subject to spatial autocorrelation. The finding that random splitting can overstate the predictive contribution of such features is not unique to crime, and the spatial block cross-validation approach used here is directly transferable to any study where training and test observations are drawn from the same geographic area.

5.6 Conclusions

This thesis demonstrates that localized violent-crime exposure contains useful predictive information for Dallas residential short-term rental pricing when evaluated under random test-set conditions, improving R^2 from 0.689 to 0.701 and reducing RMSE from 0.402 to 0.394. Hedonic regression benchmarks confirm that violent crime is negatively and significantly associated with price at all five buffer distances tested. SHAP analysis reveals that structural features remain dominant, crime variables rank among secondary predictors, and the direction of crime effects is scale-dependent: only the immediate 100-meter buffer is unambiguously negative, while broader buffers are confounded with urban centrality at the unconditional level.

The spatial block cross-validation finding is the most important methodological contribution of the thesis. The crime improvement observed under random splitting disappears under geographic holdout, indicating that spatial autocorrelation largely drives the apparent predictive value of crime features in a machine learning context. The random forest comparison confirms that this pattern is not specific to gradient boosting: a structurally different tree-based learner shows the same disappearance of the crime contribution under spatial evaluation. Researchers and practitioners should therefore interpret crime-prediction improvements cautiously when they are based on random train-test splits and should incorporate spatial cross-validation as a standard robustness check in urban prediction studies.

In Dallas, violent crime is a meaningful neighborhood disamenity, but its contribution to cross-spatial prediction generalizability is limited by the very spatial structure that makes it a theoretically interesting variable in the first place.

Taken together, the results of this thesis suggest that the method of evaluation matters as much as the method of modeling. A crime feature that improves a randomly evaluated model may not improve a spatially evaluated one. This reflects a fundamental difference between explaining prices in locations where training data already exist and predicting prices in genuinely new geographic areas. As urban prediction studies continue to expand in scope and data richness, spatial evaluation design should receive the same level of attention as feature engineering and model selection.

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